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Bitcoin Return Direction Prediction ——with Deep Learning and Interpretable Features

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ABSTRACT

Problem Definition and Research Objectives

- **Overview:** We developed a system that predict the next-day direction of Bitcoin (Single Asset) based on Deep Learning method that combines market, on-chain, macro, and sentiment features.
- **Task formulation:** Binary classification task on daily frequency.
 - ``target_t = 1[close_t > close_{t-1}]``
 - ``target_t = 0 otherwise``
- **Evaluation:** Classification metrics + unified financial backtest metrics.
 - Classification metrics (e.g., AUC/F1)
 - Financial metrics (e.g., return/Sharpe/drawdown)



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PART 1

DATA ENGINEERING

1. DATA ENGINEERING

1.1 Data Sources and Features

Source	Type	Example Variables	Role in Modeling
Binance Spot API	Market microstructure / price-volume	open, high, low, close, volume, quote_volume, trade_count	Core market state and return dynamics
Blockchain.com Charts API	On-chain network activity	active_addresses, tx_count, hash_rate, fees, difficulty	Captures blockchain usage, congestion, and mining condition
FRED	Macro-financial environment	vix, sp500_return, dxy_proxy	Adds global risk sentiment and cross-market context
Alternative.me	Crypto sentiment index	fear_greed_index	Proxies market emotion and short-term behavioral pressure

- Final dataset: 2966 rows, 27 columns
- Date range: 2018-02-02 to 2026-03-17
- Missing rate in final table: 0.00%



1. DATA ENGINEERING

1.2 Final Feature Groups

Feature Group	Variables	Why It Matters
Price & Trading	open, high, low, close, volume, quote_volume, trade_count	Captures immediate market movement and liquidity condition
On-chain Activity	active_addresses, tx_count, hash_rate, fees, difficulty	Reflects network utilization, transaction pressure, and mining regime
Macro Factors	vix, sp500_return, dxy_proxy	Encodes broader market risk and dollar/liquidity conditions
Sentiment	fear_greed_index	Adds behavioral signal from crypto crowd sentiment
Engineered Features	*Short-horizon transformed signals from price, volume, and on-chain dynamics	Transforms raw signals into model-friendly short-horizon predictors
Target	target = 1[close_t > close_{t-1}]	Binary direction label for daily prediction

- *Engineered Features Specs:

ret_1d, ret_3d, volatility_7d, ma_ratio_7_30, high_low_spread, volume_change, active_addresses_change, fees_per_tx, difficulty_change



1. DATA ENGINEERING

1.3 Leakage Control

- **R1:** Shift all modeling predictors by one period
- **R2:** Exclude the latest incomplete trading day
- **R3:** Keep a fixed target definition and test it

Implementation (from original pipeline)

```
```python
```

```
feature_columns = [...]
merged[feature_columns] =
merged[feature_columns].shift(1)
final_df = merged.dropna(subset=[*feature_columns,
"target"]).reset_index(drop=True)
```

```
```
```

Latest day exclusion: `remove\_latest\_binance\_day(...)`

Target rule: `target = (close > close.shift(1))`



PART 2

BASELINE & GRU

2. BASELINE & GRU

2.1 Baseline Experimental Results

- **Baselines: Logistic Regression (L1) and Random Forest**

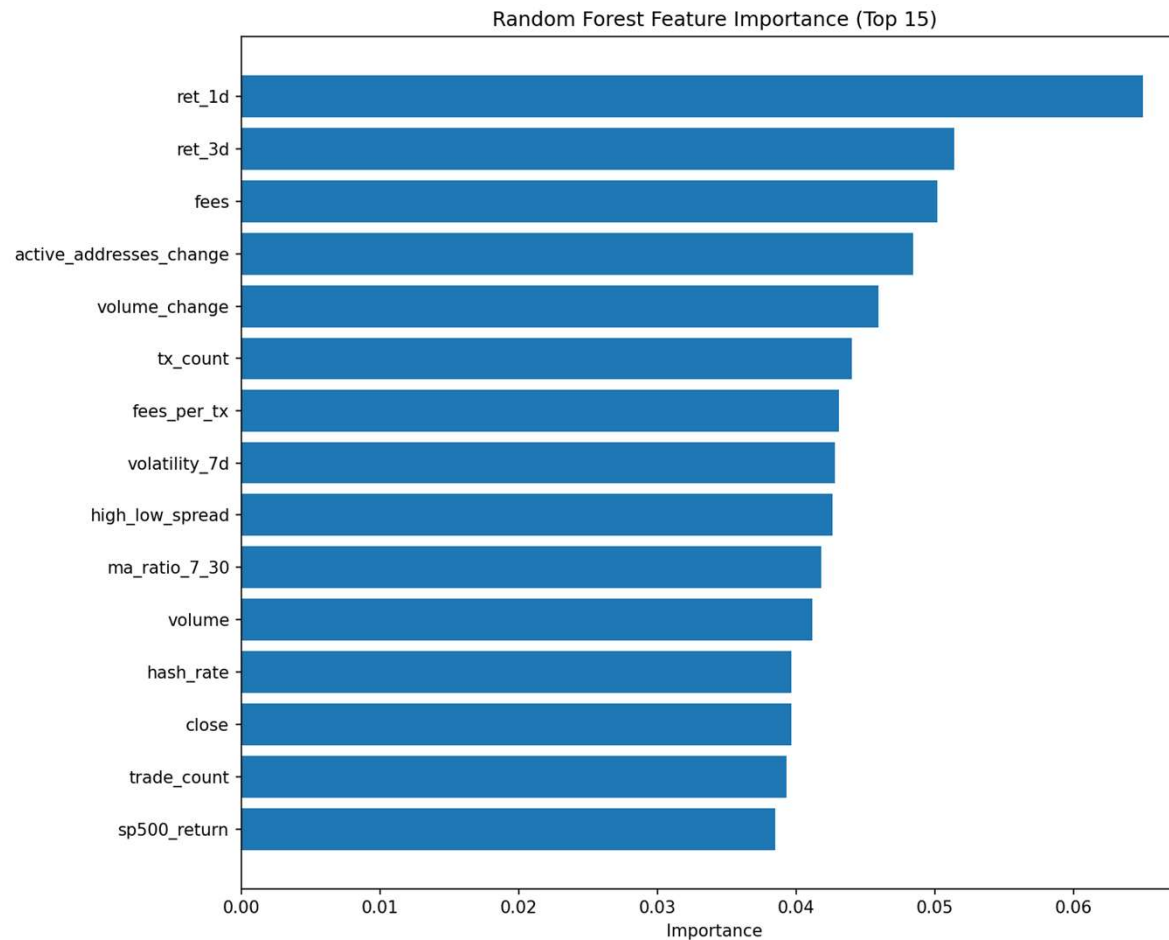
| Model | accuracy | precision | recall | f1 | auc |
|-----------------|----------|-----------|--------|-------|-------|
| Baseline-LR(L1) | 0.499 | 0.495 | 0.659 | 0.565 | 0.493 |
| Baseline-RF | 0.499 | 0.485 | 0.227 | 0.31 | 0.456 |

- Classification quality remains near-random in this run
- These models define the empirical reference level



2. BASELINE & GRU

2.1 Baseline Experimental Results



- **Random Forest importance plot shows:**
 - Short-horizon return and activity-related variables contribute most to baseline predictions;
 - While many other features have relatively limited marginal contribution.
- **Explanation:** This indicates that the baseline model is driven by a concentrated subset of signals rather than broad, uniform feature usage.



2. BASELINE & GRU

2.2 GRU Modeling and Experimental Results

- **GRU setup:** temporal split (70/15/15), class-weighted BCE
 - Key Hyperparameters: seq_len=10, hidden_size=32, lr=5e-4

| Model | accuracy | precision | recall | f1 | auc |
|-----------------|----------|-----------|--------|-------|-------|
| Baseline-LR(L1) | 0.499 | 0.495 | 0.659 | 0.565 | 0.493 |
| Baseline-RF | 0.499 | 0.485 | 0.227 | 0.31 | 0.456 |
| GRU | 0.511 | 0.556 | 0.046 | 0.084 | 0.534 |

- **Classification Signal:** GRU has the highest AUC (0.534), indicating better ranking ability.
- **Precision–Recall Trade-off:** GRU predicts “up” more conservatively: precision is higher, recall is lower.
- **Practical Reading:** Classification improvement is modest; financial impact is validated in unified backtesting.

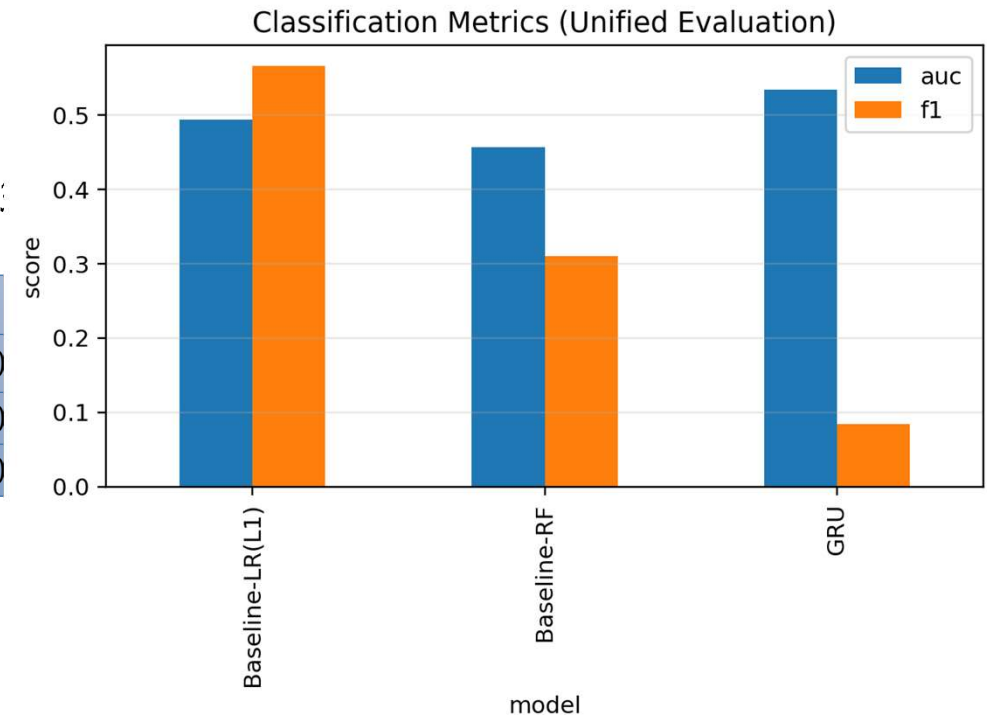


2. BASELINE & GRU

2.2 GRU Modeling and Experimental Results

- **GRU setup:** temporal split (70/15/15), class-weighted
 - Key Hyperparameters: seq_len=10, hidden_size=128

| Model | accuracy | precision | recall |
|-----------------|----------|-----------|--------|
| Baseline-LR(L1) | 0.499 | 0.495 | 0.505 |
| Baseline-RF | 0.499 | 0.485 | 0.515 |
| GRU | 0.511 | 0.556 | 0.455 |



- **Classification Signal:** GRU has the highest AUC (0.534), indicating better ranking ability.
- **Precision–Recall Trade-off:** GRU predicts “up” more conservatively: precision is higher, recall is lower.
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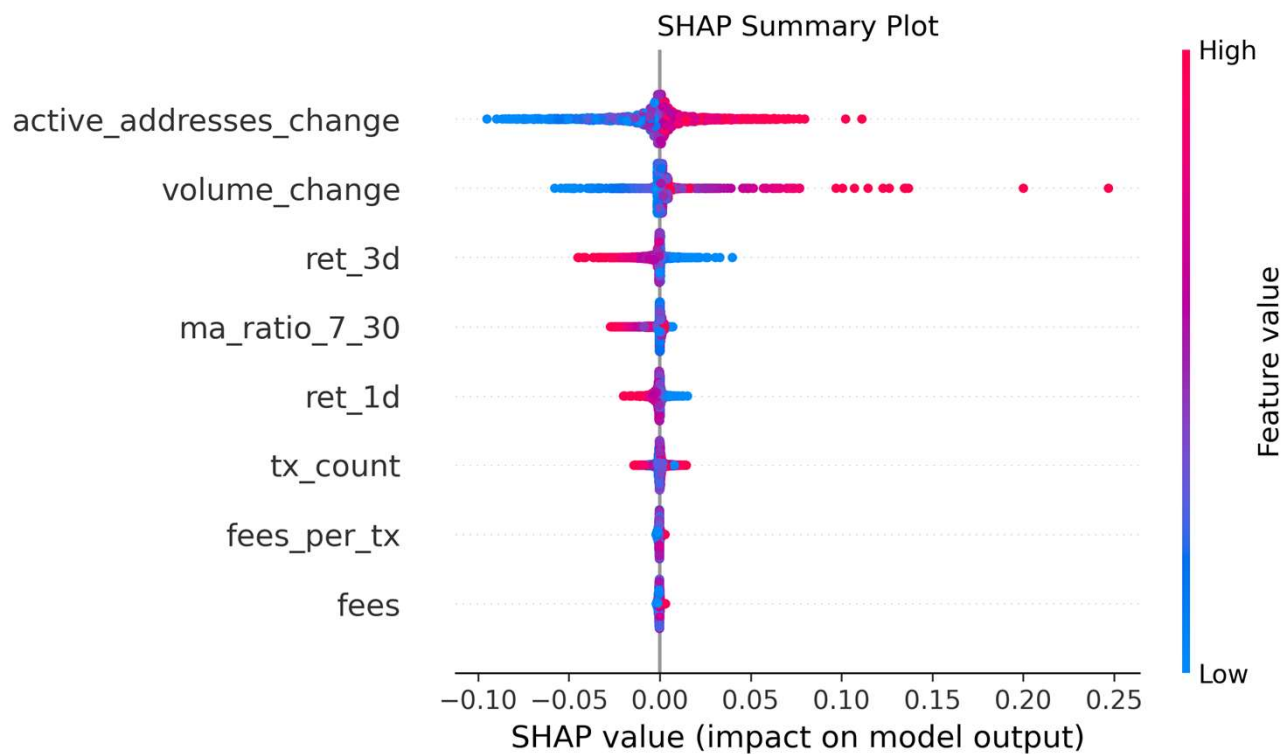


PART 3

SHAP & Financial Backtest

3. SHAP & Financial Backtest

3.1 SHAP Global Interpretation

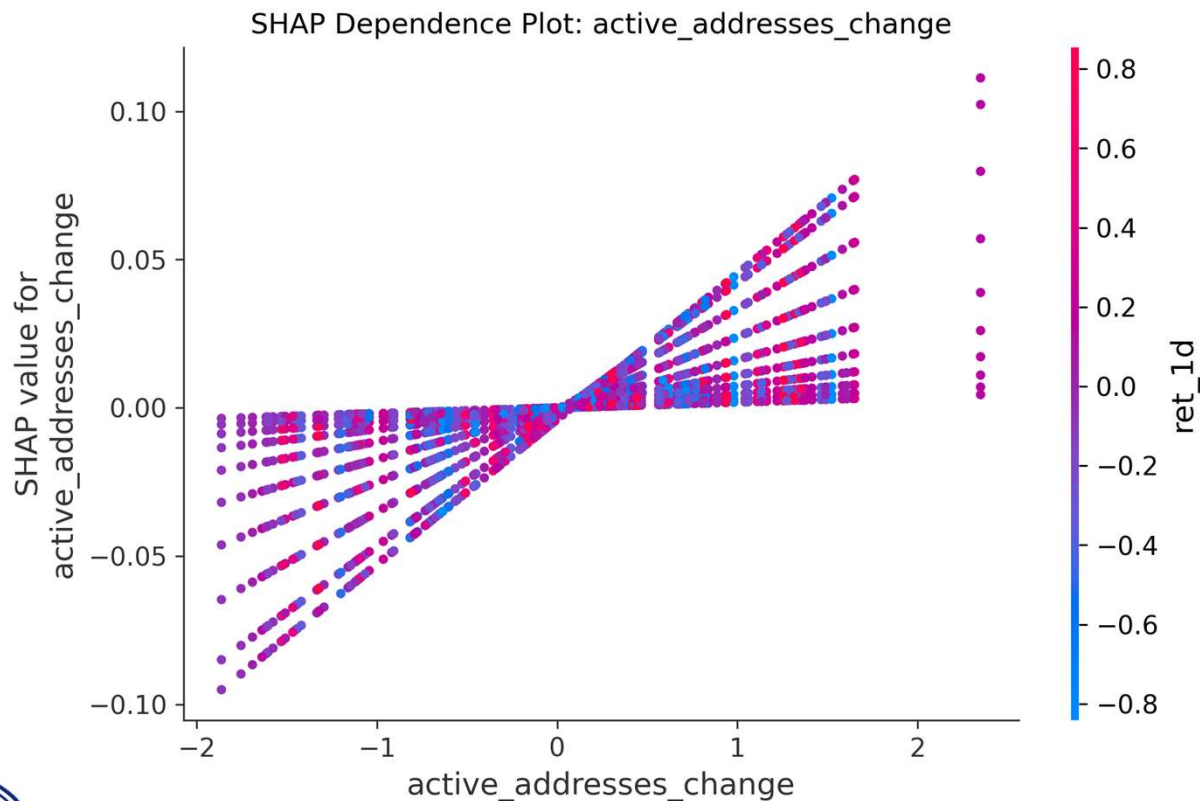


- Global SHAP ranking highlights **active_addresses_change** as the dominant signal
- **Volume** and **short-horizon** return factors are also influential



3. SHAP & Financial Backtest

3.2 SHAP Local Interpretation and Feature Ranking



| feature | mean_abs_shap |
|-------------------------|---------------|
| active_addresses_change | 0.0134 |
| volume_change | 0.00465 |
| ret_3d | 0.00364 |
| ma_ratio_7_30 | 0.00185 |
| ret_1d | 0.00171 |
| tx_count | 0.00151 |
| fees_per_tx | 0.000479 |
| fees | 0.000457 |

- Dependence analysis shows directional contribution patterns for **key on-chain activity**
- **Local behavior** is consistent with the global importance ranking



3. SHAP & Financial Backtest

3.3 Unified Backtesting Results

- **Motivation:**
 - Classification metrics alone do not measure trading consequences.
 - Backtesting translates predictions into realized return, risk, and drawdown under execution rules.
 - A unified protocol allows fair model comparison beyond accuracy-style metrics.

One protocol for all models: same signal rule, same transaction cost, same metric family.

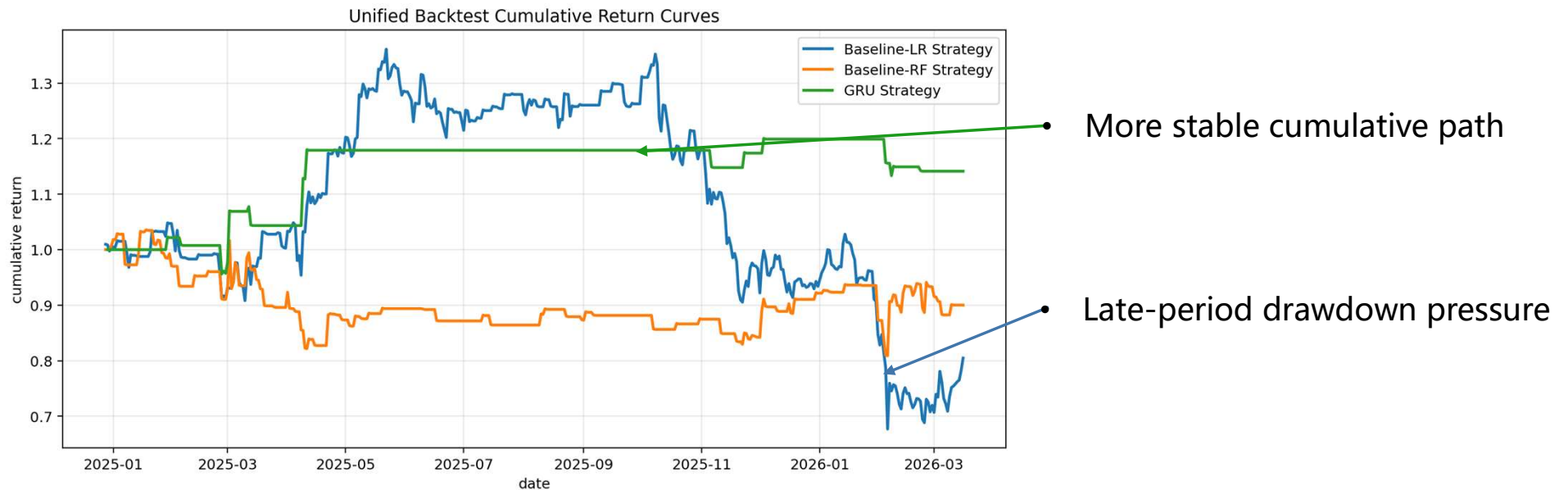
```
```python
 TRADING_FEE = 0.001
 bt["ret_next"] = bt["close"].pct_change().shift(-1)
 bt["position"] = bt["pred_up"]
 bt["trade"] = bt["position"].diff().abs().fillna(bt["position"].abs())
 bt["strategy_return"] = bt["position"] * bt["ret_next"] - bt["trade"] * TRADING_FEE
 metrics = {"total_return", "sharpe_ratio", "max_drawdown", "avg_daily_return",
 "std_daily_return", "n_days"}
```





# 3. SHAP & Financial Backtest

## 3.3 Unified Backtesting Results



- GRU remains above 1.0 for most of the test window.
- Both baseline strategies finish below 1.0.
- Performance divergence becomes clear after mid-window.

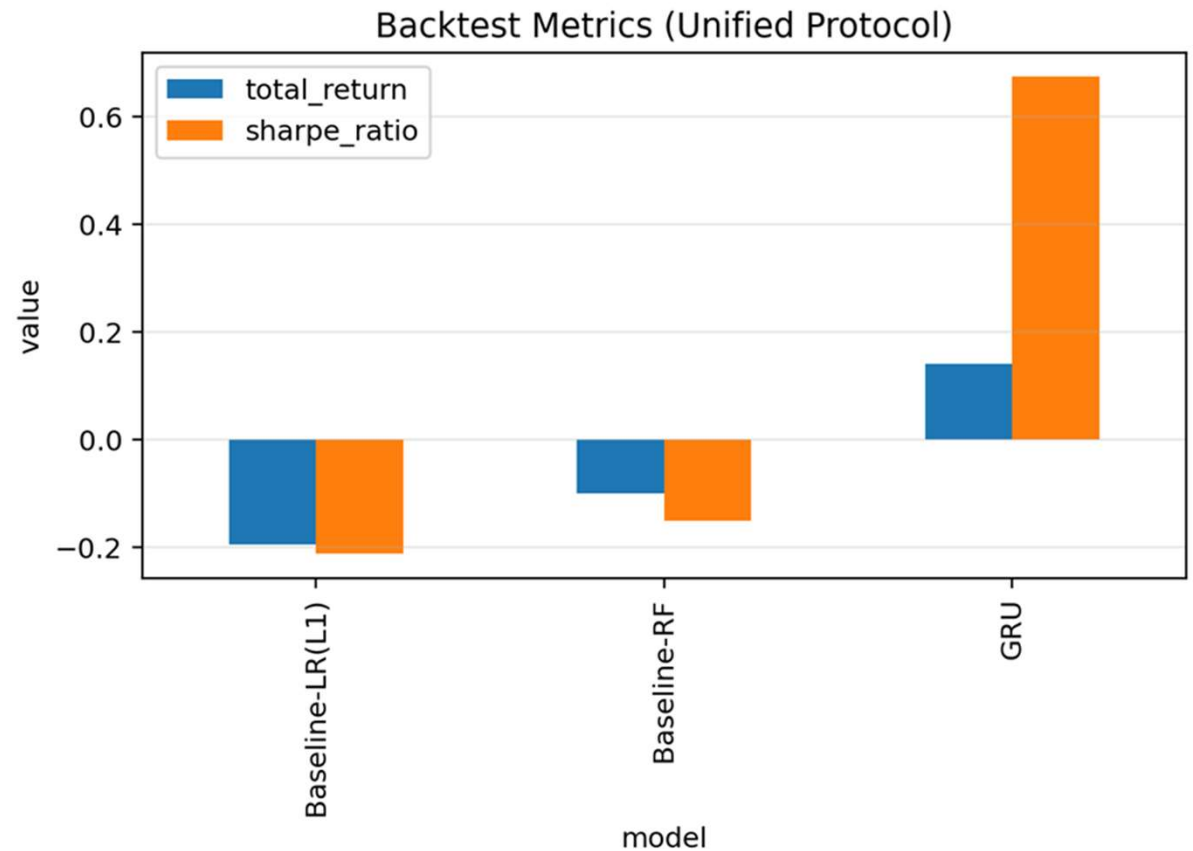


# 3. SHAP & Financial Backtest

## 3.3 Unified Backtesting Results

- Metric-level confirmation of the curve pattern (return, Sharpe, drawdown).

model	LR(L1)	RF	GRU
total_return	-0.195	-0.0998	0.141
sharpe_ratio	-0.213	-0.151	0.674
max_drawdown	-0.503	-0.219	-0.065
avg_daily_return	-0.00028	-0.00014	0.00033
std_daily_return	0.0206	0.0143	0.0077
n_days	444	444	443



PART 4

# SUMMARY

# 4. SUMMARY

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## 4.1 What We Learned

- Predictive signal exists but is **not strong enough** for aggressive alpha claims
- Interpretability adds evidence on which signals drive model decisions
- Methodological rigor (leakage control + unified evaluation) is the main contribution

## 4.2 Conclusion, Limitation and Next Steps

- **Conclusion:** Reproducible pipeline completed; GRU shows better risk-return under unified backtesting in this run;
- **Limitation:** Short-horizon BTC direction remains noisy and event-driven
- **Next Steps:** Align windows perfectly, export per-day predictions, extend to regime-aware risk targets





THANK YOU

【Designed by Group2】